

GIURGIU, Romania

WMO: 154910

Lat: 43.8753N

Lon: 25.9328E

Elev: 25

StdP: 101.03

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-9.8	-14.2	1.1	-12.5	-11.3	1.4	-9.1	14.0	-0.5	11.6	3.2	1.1	270	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	34.5	22.5	33.0	21.8	31.7	21.5	24.6	30.6	23.6	30.0	22.7	29.1	2.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.8	17.6	27.7	21.7	16.4	26.1	20.9	15.6	25.1	74.6	30.1	70.4	30.2	67.3	29.1	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	7.2	5.7	-16.5	37.8	2.5	2.0	-18.3	39.2	-19.7	40.4	-21.1	41.5	-22.9	43.0		
(5)			-16.6	26.0	2.5	2.0	-18.5	27.5	-19.9	28.6	-21.4	29.7	-23.2	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	-0.9	3.5	7.6	12.9	18.1	22.5	24.1	23.9	19.6	12.6	6.2	1.0	(6)
(7)		DBStd	9.57	4.81	5.09	4.29	4.07	3.30	2.85	2.44	2.68	3.92	4.36	4.77	4.84	(7)
(8)		HDD10.0	1077	338	186	98	20	0	0	0	0	23	132	280		(8)
(9)		HDD18.3	2676	596	416	333	165	44	4	0	2	34	182	363	538	(9)
(10)		CDD10.0	2039	0	3	23	108	252	375	436	429	287	105	20	1	(10)
(11)		CDD18.3	597	0	0	0	3	38	129	178	173	72	5	0	0	(11)
(12)		CDH23.3	6260	0	0	7	86	427	1282	1828	1836	724	69	2	0	(12)
(13)		CDH26.7	2597	0	0	1	17	110	506	815	857	277	15	0	0	(13)
(14)	Wind	WSAvg	1.9	2.0	2.2	2.4	2.3	2.0	1.7	1.6	1.5	1.8	1.8	2.1		(14)
(15)	Precipitation	PrecAvg	614	43	33	43	46	67	66	61	46	61	52	46	50	(15)
(16)		PrecMax	994	114	104	105	121	174	168	201	134	198	124	115	165	(16)
(17)		PrecMin	334	3	1	2	3	16	12	3	3	1	0	1	2	(17)
(18)		PrecStd	155	25	21	24	24	41	35	50	38	51	33	28	37	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	19.9	23.9	28.8	31.9	35.3	36.2	36.4	34.0	28.7	23.1	14.7	(19)
(20)			MCWB	8.8	13.5	13.1	18.1	21.4	23.8	23.7	22.8	20.9	19.4	15.3	10.2	(20)
(21)		2%	DB	9.5	16.3	20.5	25.9	29.4	32.7	34.4	34.3	32.0	25.0	19.1	11.9	(21)
(22)			MCWB	6.9	11.1	12.4	16.7	19.4	22.8	23.0	21.9	20.3	17.4	14.0	9.0	(22)
(23)		5%	DB	6.9	13.6	18.1	23.2	27.8	31.2	32.7	32.9	29.8	22.5	15.8	9.2	(23)
(24)			MCWB	5.2	9.5	11.3	15.5	18.8	22.2	22.4	21.2	19.5	16.4	12.6	7.3	(24)
(25)		10%	DB	4.8	11.0	15.5	20.9	25.8	29.8	31.1	31.4	27.8	20.1	13.0	6.8	(25)
(26)			MCWB	4.2	8.0	10.0	14.3	18.1	21.8	22.0	21.0	19.1	15.1	11.2	5.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	13.8	14.3	18.9	22.4	25.7	27.0	25.0	23.1	20.7	15.7	11.4	(27)
(28)			MCD B	12.1	19.0	21.8	26.0	29.9	31.8	31.7	30.5	30.9	26.4	21.6	14.4	(28)
(29)		2%	WB	7.2	11.5	12.8	17.6	20.8	24.4	25.0	23.5	21.6	18.4	14.4	9.2	(29)
(30)			MCD B	9.4	15.7	19.6	24.3	27.2	30.7	30.9	30.4	28.8	22.9	18.9	11.9	(30)
(31)		5%	WB	5.4	9.9	11.6	16.3	19.6	23.3	23.7	22.7	20.5	17.0	12.9	7.5	(31)
(32)			MCD B	6.5	13.1	17.4	21.9	25.8	29.6	29.7	29.7	27.1	21.4	15.4	9.1	(32)
(33)		10%	WB	4.1	8.3	10.4	15.1	18.6	22.4	22.8	22.0	19.6	15.7	11.2	5.7	(33)
(34)			MCD B	4.9	10.7	15.0	19.8	24.4	28.2	28.7	29.0	25.8	19.8	13.0	6.8	(34)
(35)	Mean Daily Temperature Range	MDBR	6.5	9.3	10.6	11.9	12.1	12.2	12.5	13.2	12.7	11.0	7.3	6.7	(35)	
(36)		5% DB	MCD BR	9.9	14.4	16.5	16.7	15.2	14.6	15.3	15.9	16.1	15.5	12.8	10.9	(36)
(37)			MCWBR	7.3	9.5	9.1	8.5	6.6	6.0	5.6	5.4	6.0	7.9	7.7	8.0	(37)
(38)		5% WB	MCD BR	9.1	14.2	15.6	14.8	13.8	13.4	12.9	13.7	14.2	14.0	11.8	10.4	(38)
(39)			MCWBR	6.9	9.5	8.8	7.8	6.5	6.4	5.7	5.4	5.9	7.5	7.1	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.352	0.385	0.423	0.462	0.437	0.445	0.455	0.469	0.435	0.408	0.379	0.356	(40)	
(41)		taud	2.259	2.185	2.134	2.048	2.199	2.209	2.190	2.148	2.208	2.256	2.288	2.274	(41)	
(42)		Ebn at Noon	745	784	804	803	839	830	817	789	784	751	709	699	(42)	
(43)		Edh at Noon	89	113	135	160	141	140	142	143	125	105	86	80	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.45	2.25	3.41	4.71	5.85	6.38	6.48	5.80	4.21	2.73	1.56	1.19	(44)	
(45)		RadStd	0.22	0.31	0.34	0.50	0.46	0.45	0.44	0.33	0.33	0.30	0.21	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	-119	+63	N/A

Nomenclature: See separate page